F SATISFIES THE RIM-SCHLESSINGER CONDITION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6174-6178 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 6180-6209, the complete author proof (28 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 28 of queue_artin.json (pb=1468, stmt 6174-6178, proof 6180-6209).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0GHM: sources/stacks-artin/pinned-original-source.tex, source lines 6209–6209. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07WQ: sources/stacks-artin/pinned-original-source.tex, source lines 743–746. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GH8: sources/stacks-artin/pinned-original-source.tex, source lines 5138–5153. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Situation 0GH8 the functor F satisfies the Rim-Schlessinger condition (RS).*

Proof. Recall that the condition only involves the evaluation $F(V)$ of the functor F on schemes V over S which are spectra of Artinian local rings and the restriction maps $F(V_2) \to F(V_1)$ for morphisms $V_1 \to V_2$ of schemes over S which are spectra of Artinian local rings. Thus let V/S be the spetruim of an Artinian local ring. If $\xi = (Z, u', \hat{x}) \in F(V)$ then either $Z = \emptyset$ or $Z = V$ (set theoretically). In the first

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case we see that $\hat{x}$ is a morphism from the empty formal algebraic space into W. In the second case we see that u' is a morphism from the empty scheme into X' and we see that $\hat{x} : V \to W$ is a morphism into W. We conclude that

$$F(V) = U'(V) \amalg W(V)$$

and moreover for $V_1 \to V_2$ as above the induced map $F(V_2) \to F(V_1)$ is compatible with this decomposition. Hence it suffices to prove that both U' and W satisfy the Rim-Schlessinger condition. For U' this follows from Lemma 07WQ. To see that it is true for W, we write $W = \operatorname{colim} W_n$ as in Formal Spaces, Lemma 0GHM. Say $V = \operatorname{Spec}(A)$ with $(A, \mathfrak{m})$ an Artinian local ring. Pick $n \geq 1$ such that $\mathfrak{m}^n = 0$. Then we have $W(V) = W_n(V)$. Hence we see that the Rim-Schlessinger condition for W follows from the Rim-Schlessinger condition for W_n for all n (which in turn follows from Lemma 07WQ). $\square$